

Name - Sandesh Arun Mohite

Roll No - 23

Division - T.Y(AIML)-A

Assignment - 02

Step No: 1 To Train and Save the Model

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from mpl_toolkits.mplot3d import Axes3D
```

```
df = pd.read_csv("/content/drive/MyDrive/Real estate.csv")
```

```
df.shape
```

```
(414, 8)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 414 entries, 0 to 413
Data columns (total 8 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0    No                                           414 non-null    int64
1    date                                         414 non-null    float64
2    house age                                   414 non-null    float64
3    distance to the nearest station            414 non-null    float64
4    number of convenience stores               414 non-null    int64
5    latitude                                    414 non-null    float64
6    longitude                                   414 non-null    float64
7    house price of unit area                   414 non-null    float64
dtypes: float64(6), int64(2)
memory usage: 26.0 KB
```

```
df.columns
```

```
Index(['No', 'date', 'house age', 'distance to the nearest station',
      'number of convenience stores', 'latitude', 'longitude',
      'house price of unit area'],
      dtype='object')
```

```
#Define features (X) and target (y)
X = df[['house age', 'distance to the nearest station',
      'number of convenience stores', 'latitude', 'longitude']] # independent variables
y = df['house price of unit area'] # dependent variable
```

```
X.head()
```

	house age	distance to the nearest station	number of convenience stores	latitude	longitude
0	32.0	84.87882	10	24.98298	121.54024
1	19.5	306.59470	9	24.98034	121.53951
2	13.3	561.98450	5	24.98746	121.54391
3	13.3	561.98450	5	24.98746	121.54391
4	5.0	390.56840	5	24.97937	121.54245

Next steps: [Generate code with X](#) [New interactive sheet](#)

```
#Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
#Train Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)
```

```
▼ LinearRegression ⓘ ?
LinearRegression()
```

```
#Predictions
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

```
#Evaluation
print("Coefficients (slopes):", model.coef_)
print("Intercept:", model.intercept_)
print("Mean Squared Error (MSE):", mean_squared_error(y_test, y_pred))
print("R² Score:", r2_score(y_test, y_pred))
```

```
Coefficients (slopes): [-2.70593236e-01 -4.55249601e-03  1.10512079e+00  2.36092831e+02
 -2.39036942e+01]
Intercept: -2946.6588590244446
Mean Squared Error (MSE): 54.58094520086248
R² Score: 0.6746481382828156
```

```
import pickle
```

```
pickle.dump(model, open('REPPModel.pkl', 'wb'))
```

Step No: 2 HTML Code for UI

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">

  <title>Real Estate Price Prediction</title>

  <style>

    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: Arial, Helvetica, sans-serif;
    }

    body {

      background: #f2f6fc;

      display: flex;

      justify-content: center;

      align-items: center;

      min-height: 100vh;

    }

  </style>

</head>

<body>

  <div>

    <h1>Real Estate Price Prediction</h1>

    <h2>Model Performance</h2>

    <table>

      <thead>

        <tr>

          <th>Metric</th>

          <th>Value</th>

        </tr>

      </thead>

      <tbody>

        <tr>

          <td>Coefficients (slopes)</td>

          <td>[-2.70593236e-01 -4.55249601e-03  1.10512079e+00  2.36092831e+02
 -2.39036942e+01]</td>

        </tr>

        <tr>

          <td>Intercept</td>

          <td>-2946.6588590244446</td>

        </tr>

        <tr>

          <td>Mean Squared Error (MSE)</td>

          <td>54.58094520086248</td>

        </tr>

        <tr>

          <td>R² Score</td>

          <td>0.6746481382828156</td>

        </tr>

      </tbody>

    </table>

  </div>

</body>

</html>
```

```
.container {  
  
    width: 500px;  
  
    background: white;  
  
    padding: 35px;  
  
    border-radius: 15px;  
  
    box-shadow:  
        0px 8px 25px  
        rgba(0, 0, 0, 0.15);  
  
}  
  
h1 {  
  
    text-align: center;  
  
    color: #1565c0;  
  
    margin-bottom: 8px;  
  
}  
  
h3 {  
  
    text-align: center;  
  
    color: #666;  
  
    margin-bottom: 25px;  
  
    font-weight: normal;  
  
}  
  
label {  
  
    display: block;  
  
    font-weight: bold;  
  
    margin-top: 14px;  
  
    margin-bottom: 6px;  
  
    color: #333;  
  
}  
  
input {  
  
    width: 100%;  
  
    padding: 11px;  
  
    font-size: 16px;  
  
    border: 1px solid #ccc;  
  
    border-radius: 7px;  
  
    outline: none;  
  
}  
  
input:focus {  
  
    border-color: #1565c0;  
  
}
```

```
    button {

      width: 100%;

      margin-top: 25px;

      padding: 13px;

      background: #1565c0;

      color: white;

      border: none;

      border-radius: 8px;

      font-size: 18px;

      cursor: pointer;

    }

    button:hover {

      background: #0d47a1;

    }

    .result {

      margin-top: 25px;

      padding: 18px;

      background: #e8f5e9;

      color: #2e7d32;

      border-radius: 8px;

      text-align: center;

      font-size: 20px;

      font-weight: bold;

    }

    footer {

      text-align: center;

      margin-top: 20px;

      color: #777;

      font-size: 13px;

    }

  </style>

</head>

<body>

<div class="container">

  <h1>Real Estate Price Predictor</h1>
```

```
<h3>Multiple Linear Regression</h3>

<form action="/predict" method="POST">

  <label>
    House Age
  </label>

  <input
    type="number"
    name="house_age"
    step="0.01"
    placeholder="Example: 13.3"
    required
  >

  <label>
    Distance to Nearest Station
  </label>

  <input
    type="number"
    name="distance"
    step="0.01"
    placeholder="Example: 4082.0"
    required
  >

  <label>
    Number of Convenience Stores
  </label>

  <input
    type="number"
    name="convenience_stores"
    min="0"
    step="1"
    placeholder="Example: 2"
    required
  >

  <label>
    Latitude
  </label>

  <input
    type="number"
    name="latitude"
    step="0.000001"
    placeholder="Example: 24.9830"
    required
  >

  <label>
    Longitude
  </label>

  <input
    type="number"
    name="longitude"
    step="0.000001"
    placeholder="Example: 121.5400"
    required
  >

  <button type="submit">

    Predict House Price
```

```

</button>

</form>

{% if prediction_text %}

<div class="result">

    {{ prediction_text }}

</div>

{% endif %}

<footer>

    By Dr. Sachin Balawant Takmare

</footer>

</div>

</body>

</html>

```

File ["/tmp/ipykernel_2312/2175364476.py"](#), line 34
 min-height: 100vh;
 ^
 SyntaxError: invalid decimal literal

Next steps: [Explain error](#)

Step No: 3 Flask Code for Backend

```

from flask import Flask, render_template, request
import pickle
import numpy as np

app = Flask(__name__)

# Load the trained model
model = pickle.load(open("REPPModel.pkl", "rb"))

@app.route('/')
def home():
    return render_template("index.html")

@app.route('/predict', methods=['POST'])
def predict():

    # Get input values from HTML form
    house_age = float(request.form['house_age'])

    distance = float(request.form['distance'])

    convenience_stores = float(
        request.form['convenience_stores']
    )

    latitude = float(request.form['latitude'])

    longitude = float(request.form['longitude'])

    # Arrange inputs in EXACTLY the same order
    # used while training the model
    input_data = np.array([[

```

```
        house_age,  
        distance,  
        convenience_stores,  
        latitude,  
        longitude  
    ]])  
  
    # Predict house price  
    prediction = model.predict(input_data)  
  
    # Display prediction  
    prediction_text = (  
        f"Predicted House Price : "  
        f"{prediction[0]:.2f} per unit area"  
    )  
  
    return render_template(  
        "index.html",  
        prediction_text=prediction_text  
    )  
  
if __name__ == "__main__":  
    app.run(debug=True)
```